

Security risk assessment report

Part 1: identified Vulnerabilities, three hardening tools and methods to implement

Identified Vulnerabilities:

1. The organization's employees' share passwords.
2. The admin password for the database is set to the default.
3. The firewalls do not have rules in place to filter traffic coming in and out of the network.
4. Multifactor authentication (MFA) is not used.

Hardening Tools and Methods to Implement.

I recommend implementing the following **three key hardening measures** to address the identified vulnerabilities:

1. **Password Policies** (Strongest immediate impact)
 - Enforce strong, unique passwords with complexity requirements
 - Implement password salting and hashing
 - Require regular password changes and prohibit default/reused passwords
2. **Multifactor Authentication (MFA)**
 - Enable MFA for all user accounts, especially administrative and database access
 - Use authenticator apps, hardware keys, or SMS as second factor
3. **Firewall Maintenance + Port Filtering**
 - Implement proper inbound and outbound firewall rules
 - Block unnecessary traffic and follow the principle of least privilege
 - Regularly review and update firewall configurations

Part 2: Explain your recommendations

Why these measures are effective:

- **Password Policies** directly address the risks of employees sharing passwords and the database using default credentials. Strong policies significantly reduce the success rate of brute force and credential-stuffing attacks.
- **Multifactor Authentication (MFA)** adds a critical layer of protection. Even if passwords are compromised (as happened in the recent breach), attackers would still need a second factor, greatly reducing the risk of unauthorized access.
- **Firewall Maintenance & Port Filtering** fixes the complete lack of traffic filtering rules. This prevents unauthorized access from outside the network and limits lateral movement if a breach occurs.